Website URL:

http://classwork.engr.oregonstate.edu:3212/

Feedback by the TAs and peer reviewer

Review 1 form Christian Hascall:

Overview states purpose of project: tracks and helps figure out roadtrip details and logistics. Includes facts to show importance of this in day to day living.

The database contains 6 entities, where each attribute has constraints and data types. It is in a nice clean order. I also appreciate the more in depth descriptions that provide examples of the purpose of the specific entity.

The ERD diagram is well made, clear, and concise. The PK's and FK's are labeled. And there is 1:M relationship and M:M relationships. No issues with capitalization and naming is all consistent.

I think everything looks good, and I look forwards to seeing progress!

All work is my own

Review 2 from Samson Tan:

Hi Group 49!
Reading the overview, it clearly states the purpose of the project. A database system where users can create routes, add multiple destinations, track budgets, and plan attractions at each stop. There are also specific facts listed, such as "100 users may each create multiple road trip routes" and "each route can contain 2–15 places". These numbers gives the stakeholders an idea of how big the tables are going to be, as well as the justifications for certain design decisions.
Regarding the design of the database, all 6 entities look to be in order. Each table is a single concept, which is great as there are no redundant or overloaded entities! Each entity's attributes is also well planned. TripBudget's total_budget and daily_budget being NULL makes sense if a budget hasn't been set yet. However, it's good to state this in the report to prevent any wrong assumptions. Something like

"the budget is optional until the user sets it" is good. The same goes for start_date and end_date in RoadTripRoutes. It's always good to add justifications in a report to stakeholders!
I don't see any issues with the relationships between the different entities. There is also consistency in the naming, plurality, and capitalization.
Great work, hope to see the final app!
All feedback is my original work.

Review 3 from Alysha Timmons:

Hello! This is a very good proposal, in my opinion. There is a very clear explanation about the purpose of the database, and the entities and relationships makes sense easily. In particular, I love the idea about using a junction table called RoadTripPlaces to model the many-to-many relationships and keep record of the sequence of stops for each road trip. The EDR looks nice and folows the outline neatly.
Nice job!
(all my original work)

Actions based on the feedback

The actions we took based on the feedback included improving entity descriptions to provide clearer explanations of their purposes and adding explanations for why certain attributes are allowed to be NULL, including budget attributes in the TripBudgets table and date attributes in the RoadTripRoutes table.

We also reviewed the database relationships based on the feedback. Since the reviewers confirmed that the current relationship design was appropriate, including the RoadTripPlaces intersection table for the many-to-many relationship between RoadTripRoutes and Places, we kept the overall database structure unchanged.

Upgrades to the Draft version

- Updated the entity descriptions for TripBudgets and RoadTripRoutes to explain why their optional attributes, including budget and date fields, are allowed to be NULL.

- Improved several entity descriptions by adding clearer explanations of their purposes in the road trip planning system.

- Verified the ERD documentation to ensure that the existing 1:M and M:N relationships, including the RoadTripPlaces intersection table, are clearly explained and correctly represented.

Project Step 3 Updates

The Project Step 3 implementation extends the Road Trip Guru database design into a web application interface.

The application was implemented using a React frontend and Node/Express backend and deployed on the OSU classwork server.

The website provides pages for each database entity:

- RoadTrippers
- RoadTripRoutes
- TripBudgets
- Places
- RoadTripPlaces
- Attractions

Each entity page provides interface elements for displaying records and performing CRUD operations through forms and buttons.

A Data Manipulation Language (DML) SQL file was created to support the website functionality. The DML file includes SELECT, INSERT, UPDATE, and DELETE queries for each database entity. The queries use backend-supplied variables to support record management through the website interface.

Road Trip Guru

Feifan Qi: Backend Engineer and Database Designer
Daniella Norris: Frontend Engineer and UI Designer

Road Trip Guru is a database management system designed to help road trippers organize road trips, destinations, attractions, and budgets. The system is designed for a small travel planning website where about 100 users may each create multiple road trip routes. Each route can contain 2–15 places, and each place can have several attractions such as shops, geological features, or historic sites. Users can create road trip routes, add multiple destinations to each trip, track an overall trip budget, and plan attractions to visit at each destination.

Database Structure and Entity Definitions

RoadTrippers (user table)

Represents an individual user that the road trip belongs to. A single road tripper can create multiple road trips.

- Attributes
    - road_tripper_id: int, auto_increment, PK, not NULL
    - username: varchar(255), not NULL
    - email: varchar(255), unique, not NULL
    - created_at: datetime, not NULL
- Relationships
    - RoadTrippers.road_tripper_id has a 1:M relationship with RoadTripRoutes.road_tripper_id

RoadTripRoutes

Represents a road trip route that helps users organize their planned trips, destinations, and travel information. The start_date and end_date attributes can be NULL because users may create a route before finalizing their travel schedule. Additionally, users may not want to declare these dates when creating the road trip, as they may not know the exact dates that work with their schedule until later on.

- Attributes
    - road_trip_id: int, auto_increment, unique, not NULL, PK
    - road_tripper_id: int, not NULL, FK
    - road_trip_name: varchar(255), not NULL
    - distance: int, not NULL
    - start_date: datetime, NULL
    - end_date: datetime, NULL
- Relationships
    - RoadTripRoutes has a M:N relationship with Places via RoadTripPlaces
    - RoadTripRoutes.road_trip_id has a 1:1 relationship with TripBudgets.road_trip_id

TripBudgets

Represents budget information related to a road trip route, allowing users to track overall and daily spending limits. The budget attributes can be NULL because users may create a road trip before deciding their budget.

- Attributes
    - trip_budget_id: int, auto_increment, not NULL, unique, PK
    - road_trip_id: int, not NULL, unique, FK
    - total_budget: decimal(10,2), NULL
    - daily_budget: decimal(10,2), NULL
- Relationships
    - TripBudgets.road_trip_id has a 1:1 relationship with RoadTripRoutes.road_trip_id

Places

A physical place in the world that can be added to a road trip route, such as a city or municipality.

- Attributes
    - place_id: int, auto_increment, unique, not NULL, PK
    - place_name: varchar(255), not NULL
    - place_state: varchar(255), not NULL
    - place_city: varchar(255), not NULL
- Relationships
    - Places.place_id has a 1:M relationship with Attractions.place_id
    - Places participates in M:N with RoadTripRoutes via RoadTripPlaces

RoadTripPlaces (intersection table)

Represents the association between a road trip and a place, as a place can have multiple road trips assigned to it, and a road trip can have multiple places along its route.

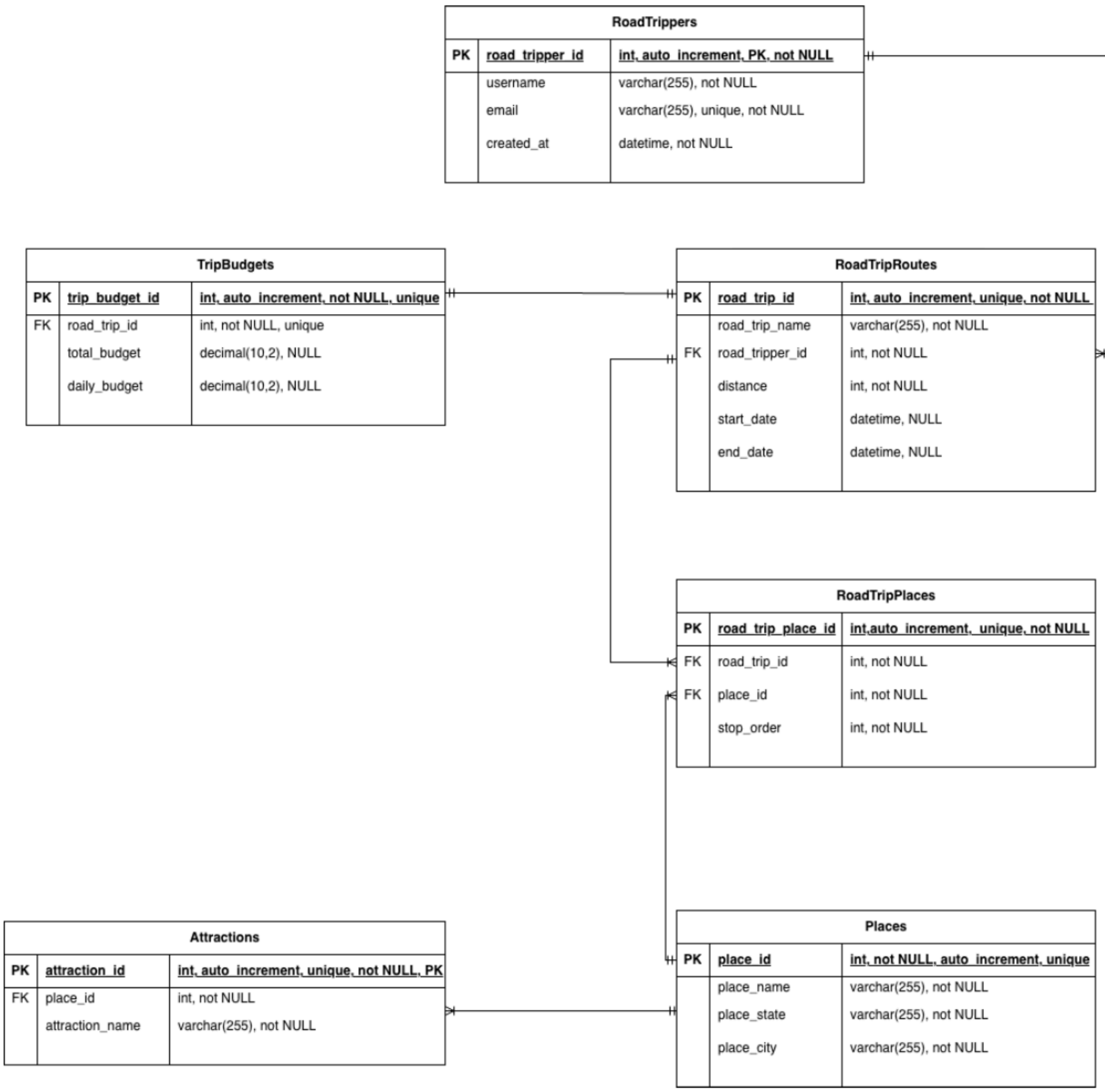
- Attributes
    - road_trip_place_id: int, auto_increment, unique, not NULL, PK
    - road_trip_id: int, not NULL, FK
    - place_id: int, not NULL, FK
    - stop_order: int, not NULL (stop_order records the order in which a place appears within a road trip route.)
- Relationships
    - RoadTripPlaces represents an intersection between RoadTripRoutes and Places, enforcing a M:N relationship

Attractions

Represents attractions (interesting shops, geological features, historic sites) that a place may contain.

- Attributes
    - attraction_id: int, auto_increment, unique, not NULL, PK

        - place_id: int, not NULL, FK
        - attraction_name: varchar(255), not NULL
- Relationships
    - Attractions.place_id has a M:1 relationship with Places.place_id



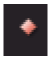
Note: This ER diagram uses crow's foot notation. "||" represents exactly one, and the crow's foot symbol represents many. For example, RoadTripRoutes and Places have a many-to-many relationship through the RoadTripPlaces intersection table.

Schema Diagram

To assist with forward-engineering our DDL Schema query, we constructed our Schema diagram using MySQL Workbench.

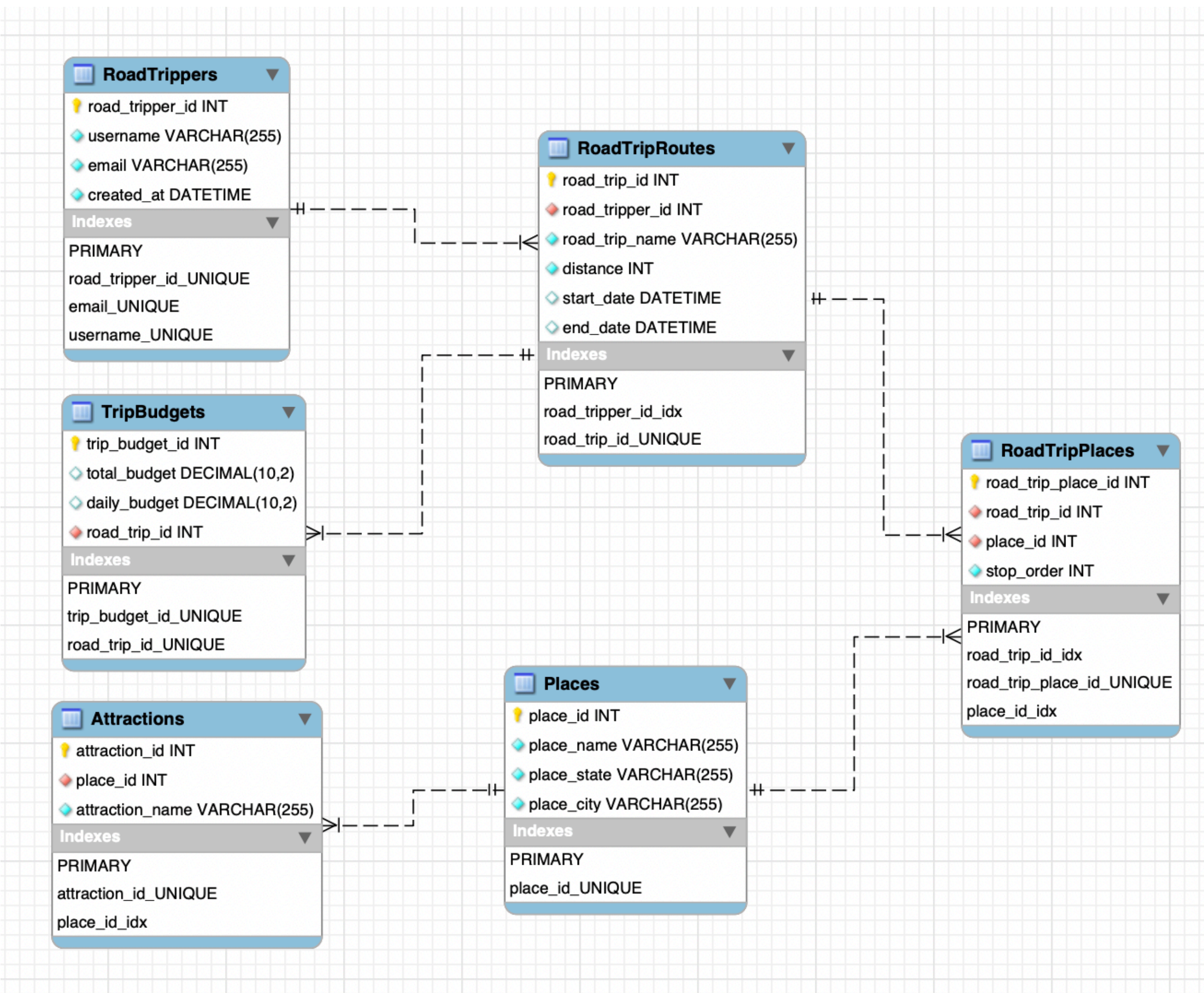
Key:

 = Primary Key

 = Foreign Key

 = NOT NULL

 = NULL



Examples (tables seeded with data)

Places

place_id	place_name	place_state	place_city
1	Crater Lake National Park	Oregon	Crater Lake
2	Golden Gate Bridge	California	San Francisco
3	Grand Canyon National Park	Arizona	Grand Canyon Village
4	Rocky Mountain National Park	Colorado	Estes Park
5	Acadia National Park	Maine	Bar Harbor
6	Mount Hood	Oregon	Government Camp
7	Yosemite National Park	California	Yosemite Valley

RoadTrippers

road_tripper_id	username	email	created_at
1	dani_travels	dani@example.com	2026-07-01 10:15:00
2	roadrunner22	roadrunner22@example.com	2026-07-02 14:30:00
3	wanderlust	wanderlust@example.com	2026-07-03 09:45:00
4	mountainlover	mountainlover@example.com	2026-07-04 16:20:00
5	coast2coast	coast2coast@example.com	2026-07-05 08:10:00

RoadTripPlaces

road_trip_place_id	road_trip_id	place_id	stop_order
1	1	6	1
2	1	1	2
3	2	2	1
4	2	7	2
5	2	1	3
6	3	3	1
7	4	4	1
8	5	5	1
9	6	2	1

RoadTripRoutes

road_trip_id	road_tripper_id	road_trip_name	distance	start_date	end_date
1	1	Pacific Northwest Adventure	1200	2026-08-01 00:00:00	2026-08-07 00:00:00
2	2	California Coast	850	2026-09-10 00:00:00	2026-09-15 00:00:00
3	3	Southwest National Parks	1600	2026-10-01 00:00:00	2026-10-10 00:00:00
4	4	Rocky Mountain Escape	950	*NULL*	2026-07-25 00:00:00
5	4	New England Fall Colors	700	*NULL*	*NULL*
6	2	Cross Country Move	2000	*NULL*	*NULL*

Attractions

attraction_id	place_id	attraction_name
1	1	Rim Drive
2	1	Wizard Island
3	2	Golden Gate Overlook
4	2	Battery Spencer
5	3	South Rim
6	3	Bright Angel Trail
7	4	Trail Ridge Road
8	4	Bear Lake
9	5	Cadillac Mountain
10	6	Timberline Lodge
11	7	Half Dome

TripBudgets

trip_budget_id	total_budget	daily_budget	road_trip_id
1	1500.00	214.29	1
2	1200.00	200.00	2
3	2500.00	250.00	3
4	1000.00	166.67	4
5	1400.00	200.00	5

Citations

All work is Group 49's own, using Canvas modules and the textbook as references. PHP myAdmin, Draw.io, and mySQL Workbench were used as diagram and schema resources.